THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 11 & 12: **The World of Algorithms, Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** the largest possible column total is 9 + 9 + 1 = 19	19 = 9 + 10 so carry is at most 1
2	**(a)** empty	we do not yet know any divisor
3	**(c)** Āryabhaṭa	Āryabhaṭīya, 499 CE
4	**(c)** A is true but R is false	work is proportional to the value, not to the number of digits
5	**(a)** 3	carries by column: [1, 1, 1, 0]
6	**(d)** all its internal angles are less than 180°	a ‘dent’ makes an angle exceed 180°
7	**(d)** parallelogram	Theorem 4 (converse of (c))
8	**(b)** bisects the third side	Theorem 7
9	**(c)** a square	equal diagonals ⇒ rhombus; perpendicular diagonals ⇒ rectangle
10	**(d)** A is false but R is true	R is true; A is false — an isosceles trapezium has a pair of equal opposite sides but is not a parallelogram

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\gcd(m, n) = \gcd(n,\ m - n)$ (1); every common divisor of m and n divides $m - n$ and conversely (1).	1+1
12	$(18, 12) \to (12, 6) \to (6, 6) \to (6, 0)$ (1); gcd $= 6$ (1).	1+1
13	True (1). Adjacent angles add up to 180° and opposite angles are equal, so each angle is 90° (1).	1+1
14	False (1). The diagonals of every rhombus are perpendicular; a rhombus with equal diagonals (or a right angle) is a square (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$825 = 375 \times 2 + 75$; $375 = 75 \times 5 + 0$ (2). gcd $= 75$, in 2 reductions (1).	2+1
16	$DE = \frac{1}{2}CA = 5$ cm, $EF = \frac{1}{2}AB = 7$ cm, $FD = \frac{1}{2}BC = 9$ cm (2). Perimeter $= 5 + 7 + 9 = 21$ cm (1).	2+1
17	$2430 = 1215 \times 2 + 0$ (2). gcd $= 1215$ (1).	2+1
	OR: $33 \bmod 7 = 5$ (1). Common divisors of 33 and 7: only 1 (1). Common divisors of 7 and 5: only 1; so the two sets of common divisors are the same, as claimed (1).	1+1+1
18	Let the diagonals meet at O; $OA = OC$ and $OB = OD$ (1). $OP = OD - DP = OB - BQ = OQ$ (1). Diagonals AC and PQ of APCQ bisect each other at O ⇒ APCQ is a parallelogram (1).	1+1+1
	OR: $AG : GD = 2 : 1$ (1). $AG = \frac{2}{3} \times 24 = 16$ cm, $GD = 8$ cm (1). $BG = \frac{2}{3} \times 30 = 20$ cm, $GE = 10$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Compute divisors(n); if the list has exactly the two entries 1 and n, report prime; else not prime (2). divisors(21) $= [1, 3, 7, 21]$ — 4 entries ⇒ not prime (1). divisors(23) $= [1, 23]$ ⇒ prime (2).	2+1+2
	OR: Compute divisors(n); keep each divisor x for which divisors(x) has exactly two entries (2). divisors(60) $= [1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60]$ (1). Prime ones: $2, 3, 5$ (2).	2+1+2
20	Let M be the midpoint of AD. In $\triangle ADC$: $MG \parallel DC$, $MG = \frac{1}{2}DC = 10$ cm; in $\triangle ABD$: $MH \parallel AB$, $MH = \frac{1}{2}AB = 6$ cm (2). As $AB \parallel DC$, both MG and MH are the line through M parallel to AB, so M, G, H are collinear (2). $GH = MG - MH = 4$ cm and $GH \parallel AB$ (1).	2+2+1
	OR: Rotate a copy of $\triangle ABC$ by 180° about the midpoint of BC: the two triangles form a parallelogram (2). Its opposite sides are equal and parallel — it is the same parallelogram tile for each pair (1). Parallelograms tile the plane by repeating the grid of two sets of parallel lines (the sticks picture) (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option

attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 11 & 12: **The World of Algorithms, Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 143	repeated reduction ends at gcd(143, 0)
2	**(b)** 5	the chain of reductions ends at gcd(3, 0)
3	**(d)** 2	each reduction replaces (m, n) by (n, m mod n); ends at gcd(1, 0)
4	**(a)** 53999	values 2, 3, …, min(m, n)
5	**(a)** 75	the rightmost element is the gcd
6	**(a)** 70°	adjacent angles of a parallelogram add up to 180°
7	**(c)** 13 cm	PQ = BC/2
8	**(c)** 12 cm	AG : GD = 2 : 1, so AG = 2/3 of AD
9	**(b)** 21 cm	AG = 2·GD, so AD = AG + GD = 3·GD
10	**(b)** 28 cm	its sides are AC/2, BD/2, AC/2, BD/2

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$60 = 16 \times 3 + 12$; $16 = 12 \times 1 + 4$; $12 = 4 \times 3 + 0$ (1). gcd $= 4$ (1).	1+1
12	$17 \bmod 5 = 2$ and $33 \bmod 7 = 5$ (1). They are the remainders on division (1).	1+1
13	The sum is 370° (1), but the angle sum of a quadrilateral is 360°, so it cannot (1).	1+1
14	$360^\circ - (85^\circ + 95^\circ + 110^\circ) = 360^\circ - 290^\circ = 70^\circ$ (2).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Examine the multiples m, $2m$, $3m$, … one by one (1). Stop at the first multiple that is divisible by n (1). Report that multiple as the lcm (1).	1+1+1
16	Let MN meet AD at X (1). $MN \parallel BC$ (Midpoint Theorem), so in $\triangle ABD$ the line MX through the midpoint M of AB is parallel to BD (1). By the converse of the Midpoint Theorem X is the midpoint of AD (1).	1+1+1
17	Start with empty list L (1). For each x in List 1: if x does not appear in List 2, append x to L (1). Report L (1).	1+1+1
	OR: Pairs: $(1, 36), (2, 18), (3, 12), (4, 9), (6, 6)$ (2). Only numbers up to $6 = \sqrt{36}$ need to be examined (1).	2+1
18	Join diagonal AC to split ABCD into $\triangle ABC$ and $\triangle ACD$ (1). Each triangle has angle sum 180° (1). The angles of the two triangles together make up the four angles of ABCD: total $= 360^\circ$ (1).	1+1+1
	OR: (i) True (1). (ii) True: the triangles on a side are congruent so the angles at the base are equal and add to 180°, i.e. 90° (1). (iii) False: a rhombus has all sides equal but need not have right angles (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$180 = 2^2 \cdot 3^2 \cdot 5$, $300 = 2^2 \cdot 3 \cdot 5^2$ (1). Store as $[(2, 2), (3, 2), (5, 1)]$ and $[(2, 2), (3, 1), (5, 2)]$ (2). For each common prime take the smaller power: $2^2, 3^1, 5^1$ (1). gcd $= 60$ (1).	1+2+1+1
	OR: (i) gcd(360, 270) (1). (ii) $360 = 270 \times 1 + 90$; $270 = 90 \times 3 + 0$ ⇒ 90 cm (1). (iii) $(360/90) \times (270/90) = 4 \times 3 = 12$ tiles (1). (iv) Division: 2 reductions; subtraction: $(360, 270) \to (270, 90) \to (90, 180) \to (180, 90) \to (90, 90) \to (90, 0)$ is 4 applications of Step 3, so 2 fewer (2).	1+1+1+2
20	(i) $\angle BCD = 180^\circ - 65^\circ = 115^\circ$ and $\angle ADC = 65^\circ$ (1). (ii) $AE = \frac{1}{2}AB = \frac{1}{2}DC = CF$ and $AE \parallel CF$ ⇒ equal parallel sides ⇒ AECF is a parallelogram (2). (iii) $AE = CF = 18$ cm (1). (iv) $2(36 + 20) = 112$ cm (1).	1+2+1+1
	OR: (i) $DE = \frac{1}{2}CA = 50$ m, $EF = \frac{1}{2}AB = 30$ m, $FD = \frac{1}{2}BC = 40$ m (1). (ii) Four triangles, all congruent to each other (each is a copy of $\triangle DEF$) (1). (iii) $50 + 30 + 40 = 120$ m (1). (iv) $60^2 + 80^2 = 3600 + 6400 = 10000 = 100^2$, so yes, right-angled at B (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET C

Chapters 11 & 12: **The World of Algorithms, Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 335	the leading 1 of 1335 is lost
2	**(d)** data structure	data structures organise information
3	**(a)** Al-Khwārizmī	Algoritmi → algorism → algorithm
4	**(a)** Both A and R are true and R is the correct explanation of A	so the units digit is written and 1 carried
5	**(d)** 2	carries by column: [1, 1, 0]
6	**(d)** intersect each other	test for convexity
7	**(b)** if the opposite angles of a quadrilateral are equal, it is a parallelogram	Theorem 3
8	**(d)** centroid	Theorem 8
9	**(b)** 360°	full turn around a vertex
10	**(b)** Both A and R are true but R is not the correct explanation of A	both true, but concurrency does not explain the 2 : 1 ratio — the Midpoint Theorem does

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Subtraction removes 2 each time (about 49 reductions) (1); division removes all the 2s at once by taking $99 \bmod 2 = 1$ (1).	1+1
12	Al-Khwārizmī (c. 820 CE) wrote on Indian numerals; the Latin form of his name, Algoritmi, gave algorismus → algorism → algorithm (2).	2
13	Pantograph (copies drawings to scale) (1); Watt's parallel-motion linkage of the steam engine (1).	1+1
14	By the converse of the Midpoint Theorem Q is the midpoint of AC (1): $AQ = QC = 9$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) $4856 + 3779 = 8635$ (1). (ii) The tens column: $5 + 7 + 1 = 13$ should give 3, but she wrote 2 (she forgot the carry 1 from $6 + 9 = 15$) (1). (iii) Step 3 (add the carry along with the digits) (1).	1+1+1
16	(i) $\angle BCD = 180^\circ - 110^\circ = 70^\circ$ (1). (ii) $2(40 + 25) = 130$ cm (1). (iii) It is still a parallelogram with the same side lengths and perimeter; only the angles change (1).	1+1+1
17	(i) $[1, 5, 25]$ (1). (ii) gcd $= 25$ (1). (iii) Lists are in increasing order, so the common list is too, and the largest common divisor is last (1).	1+1+1
	OR: (i) gcd(144, 108) (1). (ii) $144 = 108 \times 1 + 36$; $108 = 36 \times 3 + 0 \Rightarrow$ gcd $= 36$ (1). (iii) Apples $144/36 = 4$ boxes, oranges $108/36 = 3$ boxes (1).	1+1+1
18	(i) The Midpoint Theorem (1). (ii) $BC = 2 \times 35 = 70$ m (1). (iii) $PQ \parallel BC$ (1).	1+1+1
	OR: (i) A parallelogram (Varignon) (1). (ii) Sides $30, 40, 30, 40$: perimeter $= 140$ m (1). (iii) A rectangle (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) gcd(84, 126) $= 42$ (1). (ii) Common divisors $= [1, 2, 3, 6, 7, 14, 21, 42]$ (1). (iii) Boys $84/42 = 2$ rows; girls $126/42 = 3$ rows (1). (iv) The division algorithm — its number of steps depends on the digits, not the values, so it stays fast (2).	1+1+1+2
	OR: (i) One (digit + digit + carry) (1). (ii) n columns (1). (iii) $4 \times 15 = 60$ s (1). (iv) About 6000 s $= 100$ min, which is 100 times slower than 60 s (2).	1+1+1+2
20	(i) Equal diagonals ⇒ PQRS is a rhombus; perpendicular diagonals ⇒ it is also a rectangle; so it is a square (2). (ii) Each side $= 35$ m (1). (iii) Perimeter $= 140$ m; area $= 35^2 = 1225$ m^2 (1). (iv) Field area $= \frac{1}{2} \times 70 \times 70 = 2450$ m^2 (1).	2+1+1+1

	OR: (i) $2:1$, longer part towards the vertex (1). (ii) $AG = 18$, $GD = 9$, $BG = 24$, $GE = 12$ m (2). (iii) $CG = \frac{2}{3} \times 30 = 20$ m and $GF = 10$ m (1). (iv) Using two medians, G divides both in $2:1$; the third median with either gives the same point because only one point divides a segment in a given ratio (1).	1+2+1+1

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 12	repeated reduction ends at gcd(12, 0)
2	**(a)** 10	the chain of reductions ends at gcd(2, 0)
3	**(b)** 2	each reduction replaces (m, n) by (n, m mod n); ends at gcd(75, 0)
4	**(c)** 11	values 2, 3, …, min(m, n)
5	**(d)** 12	the rightmost element is the gcd
6	**(a)** 132°	adjacent angles of a parallelogram add up to 180°
7	**(d)** 7 cm	PQ = BC/2
8	**(a)** 20 cm	AG : GD = 2 : 1, so AG = 2/3 of AD
9	**(b)** 24 cm	each side of the medial triangle is half of a side
10	**(c)** 25	adjacent angles add to 180°: 6x + 30 = 180 ⇒ x = 25

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Compute divisors(n) (1). If the list has exactly two entries (1 and n), report ‘prime’; otherwise ‘not prime’ (1).	1+1
12	Store (prime, power) pairs: $[(2,2),(3,2),(5,1)]$ (2).	2
13	The diagonals bisect each other (1): $OA = 7$ cm, $OB = 5$ cm (1).	1+1
14	The tiles meeting at a point fit around it with no gap or overlap, i.e. they complete a full turn (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) $800\text{ s} \approx 13.3$ min (1). (ii) 80 000 dots (1). (iii) Only 2 more columns (1).	1+1+1
16	(i) $\frac{2}{3} \times 36 = 24$ m (1). (ii) $\frac{1}{3} \times 36 = 12$ m (1). (iii) The Centroid Theorem — medians are concurrent and divide in the ratio 2 : 1 (1).	1+1+1
17	(i) 6 (1). (ii) The number of divisors of 12 (1). (iii) j = 1, 2, 3, 4, 6, 12 (1).	1+1+1
	OR: (i) $(7,3) \to (3,4) \to (4,3) \to (3,1) \to (1,2) \to (2,1) \to (1,1) \to (1,0)$ i.e. 5 applications of Step 3 (1). (ii) $(7,3) \to (3,1) \to (1,0)$: 2 reductions (1). (iii) Division is shorter by 3 reductions (1).	1+1+1
18	(i) $70 + 80 + 100 + 110 = 360°$ (1). (ii) $360°$ — a full turn, no gap or overlap (1). (iii) Yes, the arrangement can be repeated (rotate a copy by $180°$ about the midpoint of a side) so any quadrilateral tiles the plane (1).	1+1+1
	OR: (i) A rectangle (the diagonals are perpendicular) (1). (ii) EF $= GH = \frac{1}{2}AC = 12$ cm; FG $= HE = \frac{1}{2}BD = 5$ cm (1). (iii) $2(12 + 5) = 34$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $k = 2$ ✓, $k = 5$ ✓, $k = 10$ ✓ (others up to 20 ×) (1). (ii) Final value $10 = \gcd(20, 30)$ (1). (iii) $k = 2, \ldots, 20$: 19 values (1). (iv) Use $\gcd(m,n) = \gcd(n, m \bmod n)$: $30 = 20 \times 1 + 10$; $20 = 10 \times 2 + 0$ gives 10 in 2 steps (2).	1+1+1+2
	OR: (i) $(11,2) \to (2,9) \to (9,2) \to (2,7) \to (7,2) \to (2,5) \to (5,2) \to (2,3) \to (3,2) \to (2,1) \to (1,1) \to (1,0)$: Step 3 is applied 7 times (2). (ii) $(11,2) \to (2,1) \to (1,0)$ (1). (iii) 7 versus 2 (1). (iv) Subtraction depends on the value, the division algorithm does not grow with the value (1).	2+1+1+1
20	(i) $40 + 50 + 30 + 240 = 360°$ (1). (ii) $360°$ (1). (iii) Rotate a copy through $180°$ about the midpoint of each edge of every copy added; the copies fit exactly (2). (iv) Each angle of a regular pentagon is $108°$; 360 is not a multiple of 108, so copies cannot fill a full turn around a point (1).	1+1+2+1
	OR: Adjacent angles add up to $180°$, so half of $\angle A$ plus half of $\angle B$ is $90°$ (1). In the triangle formed by AB and the bisectors of $\angle A$ and $\angle B$, the third angle is $180° - 90° = 90°$ (2). The same holds at each of the other three corners, so the figure PQRS formed has four right angles: a rectangle (1). If $AB = BC$ (a rhombus), the bisectors are the diagonals and all meet in one point, so no quadrilateral is formed (1).	1+2+1+1

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